Practical No. 1

Aim: Create a new user & grant privileges of creating session, table & view to this new user also check connect or not.

-- Create user

SQL> create user Prerna identified by prerna123;

User created.

-- Grant privileges to user

SQL> grant create table, create session, create view to Prerna;

Grant succeeded

-- Connect user

SQL> CONNECT Prerna/prerna123;

Connected.

Practical No. 2

Aim: Connect to the new user, assign a tablespace, verify it works, create a table, and insert data.

```
-- Tablespace privilege
SQL> ALTER USER Prerna
  2     DEFAULT TABLESPACE users
  3     TEMPORARY TABLESPACE temp
  4     QUOTA 10M ON users;
User altered.
-- Connect user
SQL> conn
Enter user-name: Prerna
Enter password:
Connected.
-- Create table
SQL> CREATE TABLE employees (
  2     emp_id NUMBER PRIMARY KEY,
  3     emp_name VARCHAR2(50),
  4     salary NUMBER(10,2)
  5  );
Table created.
-- Insert data in table
SQL> INSERT INTO employees VALUES (1, 'John Doe', 50000);
1 row created.
```

```
SQL> INSERT INTO employees VALUES (2, 'Jane Smith', 60000);
1 row created.
-- Show data
SQL> SELECT * FROM employees;
   EMP_ID EMP_NAME                                           SALARY
---------- -------------------------------------------------- ----------
         1 John Doe                                           50000
         2 Jane Smith                                         60000
```

Practical No. 3

Aim: Create a new role named manager for with same privileges. Grant this role new user. Then change the password of newly created user.

-- Create role

SQL> create role manager;

Role created.

-- Grant privileges to role

SQL> grant create table, create view to manager;

Grant succeeded.

-- Assign role to user

SQL> grant manager to Prerna;

Grant succeeded.

-- Chage password

SQL> password

Changing password for PRERNA

Old password:

New password:

Retype new password:

Password changed

Practical No. 4

Aim: Revoke the privileges from role manager. Revoke role from user then delete that user.

-- Revoke privileges

SQL> Revoke create table, create view from manager;

Revoke succeeded.

-- Revoke role from user

SQL> Revoke manager from Prerna;

Revoke succeeded.

-- Delete user

SQL> drop user Prerna;

User dropped.

Practical No. 5

Aim: Create a tablespace named “Std_student” with file name ‘Stu_datafile.dbf” having size of 10mb.

```
-- Grant privilege

SQL> GRANT CREATE TABLESPACE TO Prerna;

Grant succeeded.

-- Create tablespace

SQL> CREATE TABLESPACE Std_student
  2  DATAFILE 'Stu_datafile.dbf'
  3  SIZE 10M;

Tablespace created.
```

Practical No. 6

Aim: Create a tablespace named “ts_mydemo” with two datafile named “ts_mydemo01.dbf” & “ts_mydemo02.dbf “having size of 50mb and 64mb respectively. Make the tablespace auto extendable and limit auto extending to size of 2048mb.In auto extending, next extend should of 32mb.

-- Create two tablespace

```
SQL> CREATE TABLESPACE ts_mydemo
  2  DATAFILE 'mydemo01.dbf' SIZE 50M
    AUTOEXTEND ON NEXT 32M
    MAXSIZE 2048M,
  3          'mydemo02.dbf' SIZE 64M
    AUTOEXTEND ON NEXT 32M
    MAXSIZE 2048M;

    Tablespace created.
```

Practical No. 7

Aim: Alter the last datafile "ts_mydemo01.dbf" so that its size will automatically extend by 5mb& limit its size to 500mb.

```
SQL> SELECT file_id, file_name
  2  FROM dba_data_files
  3  WHERE tablespace_name = 'TS_MYDEMO';
   FILE_ID    FILE_NAME
----------
        10
C:\ORACLEXE\APP\ORACLE\PRODUCT\11.2.0\SERVER\DATABASE\MYDEMO01.
DBF
        11
C:\ORACLEXE\APP\ORACLE\PRODUCT\11.2.0\SERVER\DATABASE\MYDEMO02.
DBF
-- Alter Datafile
SQL> ALTER DATABASE
2  DATAFILE
'C:\ORACLEXE\APP\ORACLE\PRODUCT\11.2.0\SERVER\DATABASE\
MYDEMO01.DBF'
3  AUTOEXTEND ON
4  NEXT 5M
5  MAXSIZE 500M;
Database altered.
```

Practical No. 8

Aim: Create temporary table space with size of 5mb which will be automatically increase on demand.

-- create tablespace

SQL> create temporary tablespace temp_demo

2 temp file 'temp01.dbf' size 5m auto extend on;

Tablespace created.

Practical No. 9

Aim: Add new Redo log file of size 10mb to existing database.

SQL> alter database add logfile 'c:/program files/oracle/log5.ora' size 10m;

Database altered.

Practical No. 10

Aim: Display the list of all Redo log files and control files.

SQL> select * from v$logfile;

```
   GROUP# STATUS  TYPE
---------- ------- -------
MEMBER
--------------------------------------------------------------------------------------------------------------------------------
IS_
---
         2        ONLINE
C:\ORACLEXE\APP\ORACLE\FLASH_RECOVERY_AREA\XE\ONLINELOG\O1_MF_2_MM8H1MLH_.LO
G
YES

         1 STALE   ONLINE
C:\ORACLEXE\APP\ORACLE\FLASH_RECOVERY_AREA\XE\ONLINELOG\O1_MF_1_MM8H1MHF_.LO
```

SQL> select * from v$controlfile;

Practical No. 11

Aim: Display the list, datafiles, tablespace names & size in MB.

```
SQL> clear breaks;
breaks cleared
SQL> set linesize 130
SQL> set pagesize 60
SQL> break on tablespace_name skip 1
SQL> col tablespace_name format a15
SQL> col file_name format a50
SQL> col tablespace_kb heading 'TABLESPACE|TOTAL_KB'
SQL> col kbytes_free heading 'TOTAL FREE|KBYTES'
SQL> select file_name,tablespace_name, ROUND(bytes/1024000) mb
  2  from dba_data_files order by 1;
```

FILE_NAME	TABLESPACE_NAME	MB
C:\ORACLEXE\ORADATA\DATAFILE.DBF	TESTING	10
C:\ORACLEXE\ORADATA\TS_MYDEMO01.DBF	TS_MYDEMO	51
C:\ORACLEXE\ORADATA\TS_MYDEMO02.DBF		66
C:\ORACLEXE\ORADATA\XE\DATAFILE.DBF	DBA_TABLESPACE	10
C:\ORACLEXE\ORADATA\XE\SYSAUX.DBF	SYSAUX	451
C:\ORACLEXE\ORADATA\XE\SYSTEM.DBF	SYSTEM	348
C:\ORACLEXE\ORADATA\XE\UNDO.DBF	UNDO	92
C:\ORACLEXE\ORADATA\XE\USERS.DBF	USERS	102

```
8  rows selected.
```

Practical no. 12

Aim: Create rollback segment in separate tablespace with proper storage option.

```
1  Create tablespace small_rollbacks
2  datafile 'c:program files/oracle/small_rollbacks.dbf' size 20m reuse
  3* default storage(initial 250k next 250k maxextents 20)

Tablespace created.
SQL> alter session set undo_suppress_errors=true;

Session altered.
SQL>  create  rollback segment small_rollbacks
2  tablespace small_rollbacks
3  storage(initial 250k next 250k maxextents 20);

Rollback segment created.
```

Practical no. 13

Aim: Create a 3 new abstract datatype as given below

1. NAME_TYPE(F_NAME,M_NAME,L_NAME)

2. POST_TYPE(MAIN,SFX)

3. ADDR_TYPE(STREET,CITY,STATE,POSTCODE)

```
  SQL> create type name_type as object(f_namevarchar(15),
2  m_init char(1),
  3 l_namevarchar2(20));

Type created.
SQL> create type post_type as object(main integer,sfx integer);

Type created.
SQL> create type add_type as object(
2  street varchar2(35),
3  cityvarchar(15),state char(2),postcode post_type);

Type created.
```

Practical no. 14

Aim: Create a table studinfo having column name,rollno,phy,maths,comp,total and percentage. Write a trigger to autometicallycalculate students total marks and percentage of students while insertion.

```
SQL> create table studinfo
2  (name varchar2(20),
3  rollno number(3),
4  phy  number(3),
5  math  number(3),
6  comp number(3),
7  total  number(5),
8  per number(3,2));
Table created.

SQL> create or replace trigger totper
2  before insert on studinfo
3  for each row
4  begin
5  :new.total:=:new.math+:new.phy+:new.comp;
6  :new.per:=(:new.total/300)*100;
7 endtotper;
8 /
Trigger created.
  SQL>insert into studinfo(name,rollno,phy,math,comp)
2  values('sid',1,7,7,7)/
```

1 row created.

SQL> select * from studinfo;

NAME	ROLLNO	PHY	MATH	COMP	TOTAL	PER
sid	1	7	7	7	21	7

Practical no. 15

Aim: Write a SQL query to display the following data information

.1)Name of database

2)Database startup time

3)Database size information(database size,used space, free Space)

```
SQL> select name from v$database;

NAME
---------
ORCL
SQL>select to_char(startup_time,'hh24:mi dd-mm-yy')"startup time"
2 fromv$instance;
startup time
----------------
11:10 06-10-18

1  select round(sum(used.bytes)/1024/1024/1024)|| 'gb' "database size",
  2   round(sum(used.bytes)/1024/1024) || 'mb' "used space",
3  round(free.p/1024/1024) || 'mb' "free space"
4  from(select bytes
5  fromv$datafile
6   union all
7  select bytes
8   from v$tempfile
9  union all
10  select bytes
```

```
11  fromv$log) used,
12  (select sum(bytes)as p
13  fromdba_free_space)free
 14* group by free.p
SQL> /

database size        used space           free space
---------------------- ---------------------- ----------------------
1gb                  1104mb               186mb
```